

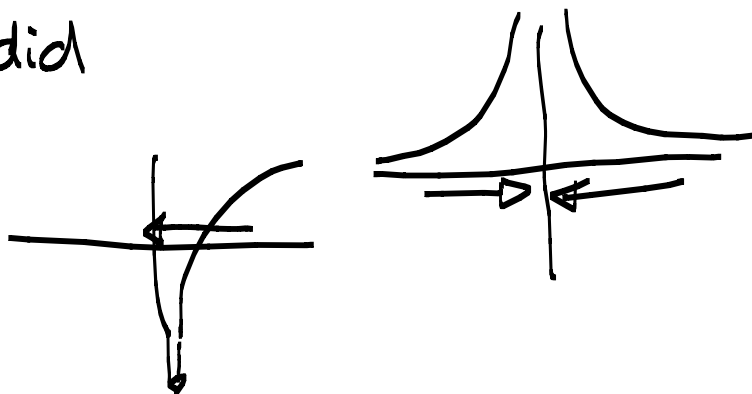
Unit 2.5 limits at infinity and horizontal asymptotes pp 130-143

In unit 2.2 we did

$$\lim_{x \rightarrow 0} \frac{1}{x^2} \quad \left(\frac{\infty}{0} \right)$$

$$= \infty \quad (\text{sign})$$

$$\lim_{x \rightarrow 0^+} \ln x = -\infty$$

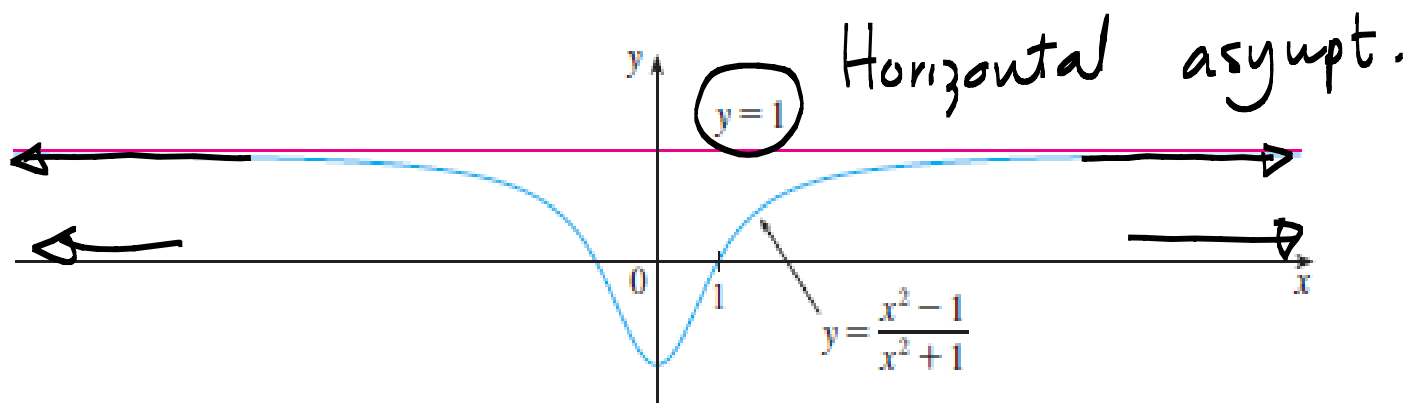


x	$f(x)$
0	-1
± 1	0
± 2	0.600000
± 3	0.800000
± 4	0.882353
± 5	0.923077
± 10	0.980198
± 50	0.999200
± 100	0.999800
± 1000	0.999998

What about

$$\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^2 + 1} \quad ?$$

$$= 1 \quad ??$$

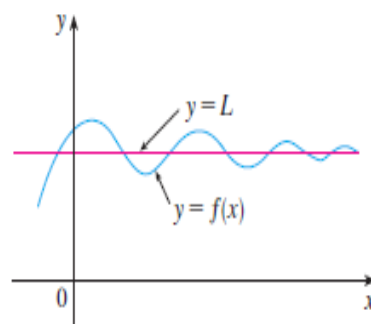
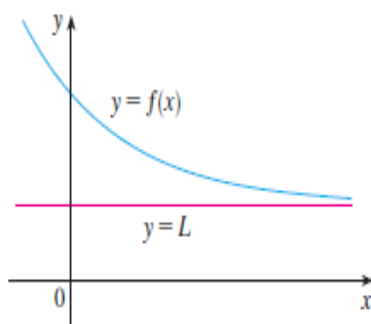
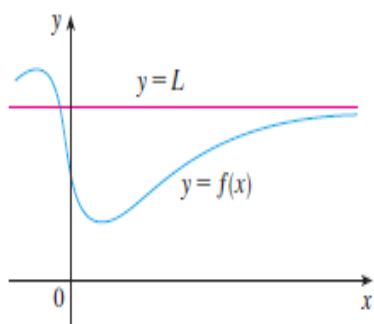


Definition 1 : p130

Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large. $x \rightarrow \infty \Rightarrow f(x) \rightarrow L$



Definition 2 SELF $x \rightarrow -\infty$.

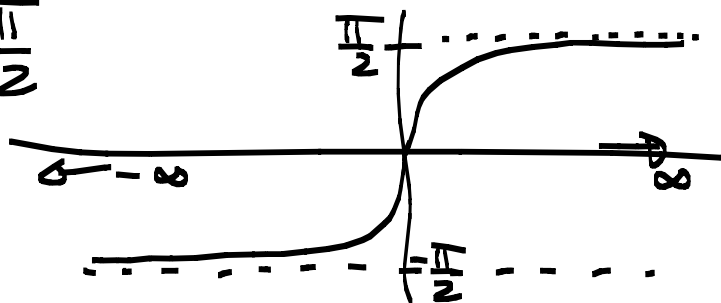
Definition 3

The line $y = L$ is a horizontal asymptote of $y = f(x)$ if either

$$\lim_{x \rightarrow -\infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow +\infty} f(x) = L$$

Example

① $\lim_{x \rightarrow -\infty} \tan^{-1} x = -\frac{\pi}{2}$



② Example 1 p 132 Find the infinite limits, limits at infinity and asymptotes of f .

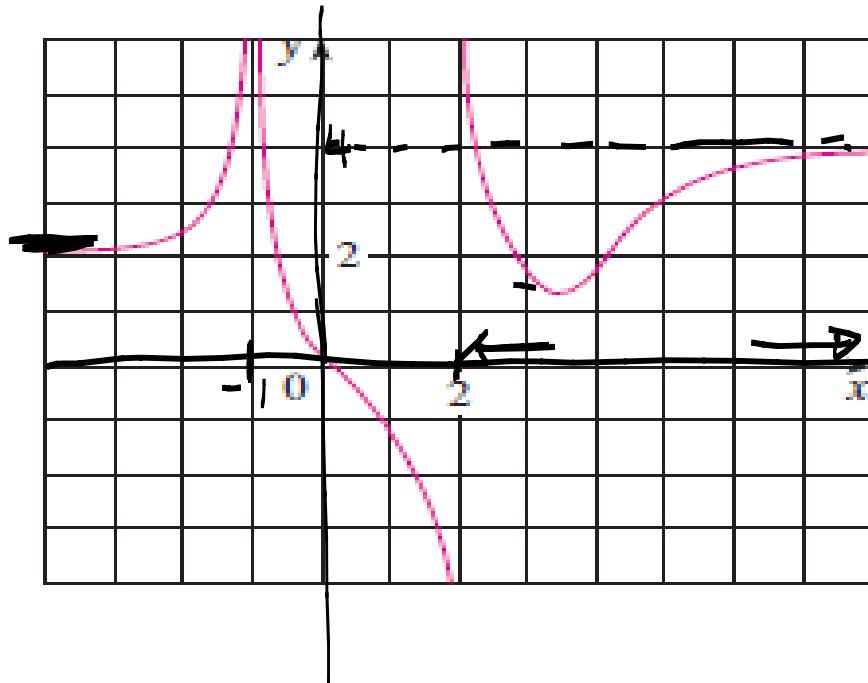
$$\lim_{x \rightarrow -1} f = \infty$$

$$\lim_{x \rightarrow 2^-} f = -\infty$$

$$\lim_{x \rightarrow 2^+} f = \infty$$

V. A at

$$x = -1 \text{ and } x = 2$$



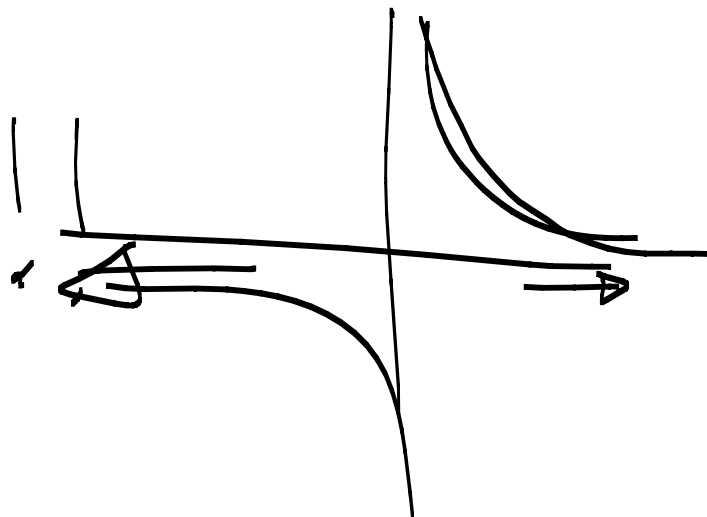
$$\lim_{x \rightarrow \infty} f = 2$$

$$\lim_{x \rightarrow -\infty} f = 2$$

H. A at

$$y = 2 \text{ and } y = 4$$

③ $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$



This example leads to:

Theorem 5 p 133 NB

If $r > 0$ a rational number then

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0, \quad \lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$$

With all rational functions we know

What $\lim_{x \rightarrow \infty} \frac{Q(x)}{P(x)}$ will be!

$$\begin{aligned} \textcircled{1} \quad & \lim_{x \rightarrow \infty} \frac{2x-1}{x+3} \\ &= \lim_{x \rightarrow \infty} \frac{x(2 - \frac{1}{x})}{x(1 + \frac{3}{x})} \end{aligned}$$

$$\left(\frac{\infty}{\infty}\right)$$

degrees equal:
answer a real number

$$= 2$$

degree on top > degree at bottom
 ∞

$$\begin{aligned} \textcircled{2} \quad & \lim_{x \rightarrow \infty} \frac{x^2 + x - 1}{2x - 5} \\ &= \lim_{x \rightarrow \infty} \frac{\cancel{x^2}(1 + \frac{1}{x} - \frac{1}{x^2})}{\cancel{x^2}(\frac{2}{x} - \frac{5}{x^2})} \end{aligned}$$

$$\left(\frac{\infty}{0}\right)$$

$$= \infty \quad (\text{Sign!!!})$$

$$\begin{aligned} \textcircled{3} \quad & \lim_{x \rightarrow \infty} \frac{3x^2 + x + 2}{x^3 + 1} \\ &= \lim_{x \rightarrow \infty} \frac{\cancel{x^3}(\frac{3}{x} + \frac{1}{x^2} + \frac{2}{x^3})}{\cancel{x^3}(1 + \frac{1}{x^3})} \end{aligned}$$

degree on top < degree at bottom.
0

$$x^3(1 + 1/x^3)$$

$$= 0$$

④ Remember! * $\boxed{\sqrt{x^2} = |x|}$

Find the H.A. if any of $\sqrt{2 \cdot 2} = \sqrt{2} \sqrt{2}$

$$f(x) = \frac{\sqrt{9x^2 - 4}}{4 - 8x}$$

$$= \frac{\sqrt{x^2(9 - \frac{4}{x^2})}}{4 - 8x} = \frac{\sqrt{x^2} \sqrt{9 - \frac{4}{x^2}}}{x(\frac{4}{x} - 8)}$$

look at the difference ($-\infty$ or ∞)

$$\lim_{x \rightarrow \infty} \frac{|x| \sqrt{9 - \frac{4}{x^2}}}{x(\frac{4}{x} - 8)}$$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{x} \sqrt{9 - (\frac{4}{x^2})}}{\cancel{x}(\frac{4}{x} - 8)}$$

$$= \frac{3}{-8}$$

$$\lim_{x \rightarrow -\infty} \frac{|x| \sqrt{9 - \frac{4}{x^2}}}{x(\frac{4}{x} - 8)}$$

$$\lim_{x \rightarrow -\infty} \frac{-\cancel{x} \sqrt{9 - \frac{4}{x^2}}}{\cancel{x}(\frac{4}{x} - 8)}$$

$$= \frac{-3}{-8} = \frac{3}{8}$$

$\therefore y = -\frac{3}{8}$ and $y = \frac{3}{8}$ are H.A

To be continued.

Unit 2.5 limits at infinity and horizontal asymptotes pp 130-143

Note Title

2016/04/06

Continued.

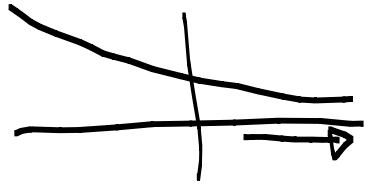
$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 4x} - x \right) \left(\frac{\sqrt{x^2 + 4x} + x}{\sqrt{x^2 + 4x} + x} \right) (\infty - \infty) \\ &= \lim_{x \rightarrow \infty} \frac{x^2 + 4x - x^2}{\sqrt{x^2 \left(1 + \frac{4}{x} \right)} + x} \\ &= \lim_{x \rightarrow \infty} \frac{4x}{x \left(\sqrt{1 + \frac{4}{x}} + 1 \right)} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \sqrt{x^2} &= |x| \\ \text{if } x &\rightarrow \infty \\ |x| &= x \end{aligned}$$

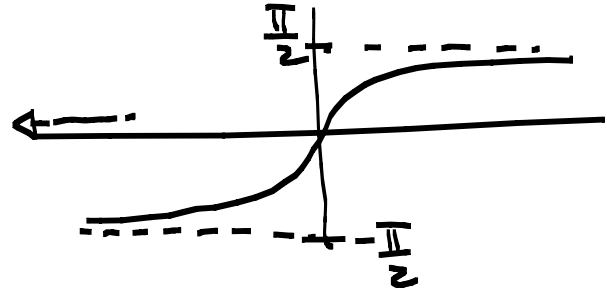
Continued:

Use graphs to find the limits of:

$$\textcircled{1} \lim_{x \rightarrow \infty} e^{-3x} = 0$$



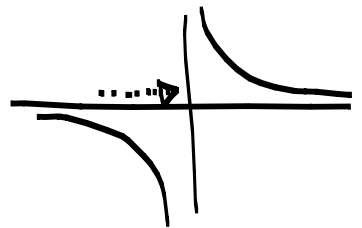
$$\textcircled{2} \lim_{x \rightarrow -\infty} \arctan x = -\frac{\pi}{2}$$



$$\textcircled{3} \lim_{x \rightarrow -\infty} \arctan(\underbrace{-2x^3}_{\infty}) = \frac{\pi}{2}$$

$$\lim_{x \rightarrow -\infty} (-2x^3) = \infty$$

$$\frac{1}{x} \xrightarrow{x \rightarrow 0^-} -\infty$$



$$\textcircled{4} \lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0$$

$$\textcircled{5} \lim_{x \rightarrow -\infty} e^x = 0$$

$$\textcircled{6} \lim_{x \rightarrow \infty} \frac{2e^x}{e^x - 5} = \lim_{x \rightarrow \infty} \frac{\cancel{e^x} (2)}{\cancel{e^x} (1 - \frac{5}{\cancel{e^x}})} = 2$$

Determine all the asymptotes
(Vertical and Horizontal) of

$$f(x) = \frac{x^3}{x^2-1} = \frac{x^3}{(x-1)(x+1)}$$

V.A Possible where $x=1$ or $x=-1$

$$\left. \begin{array}{l} \lim_{x \rightarrow 1^+} \frac{x^3 \oplus}{(x-1)_+ (x+1)_+} = +\infty \\ \lim_{x \rightarrow 1^-} \frac{x^3 \oplus}{(x-1)_- (x+1)_+} = -\infty \end{array} \right\} \Rightarrow x=1 \text{ is a V.A.}$$

$$\left. \begin{array}{l} \lim_{x \rightarrow -1^+} \frac{x^3 \ominus}{(x-1)_- (x+1)_+} = +\infty \\ \lim_{x \rightarrow -1^-} \frac{x^3 \oplus}{(x-1)_- (x+1)_-} = -\infty \end{array} \right\} \Rightarrow x=-1 \text{ is a V.A.}$$

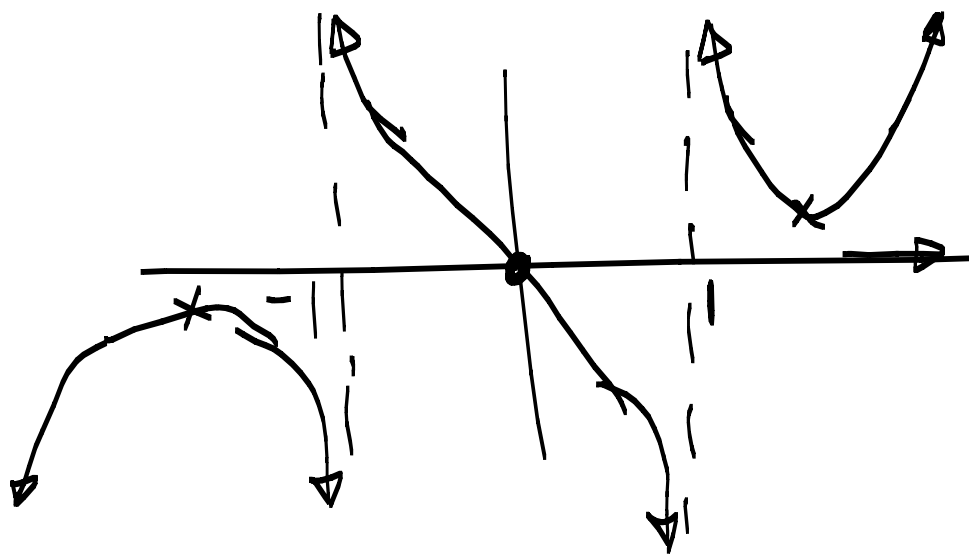
H.A. $\lim_{x \rightarrow \infty} \frac{x^3}{x^2-1} = +\infty$

$$\left[\frac{\cancel{x^3}(1)}{\cancel{x^3}(\frac{1}{x} - \frac{1}{x^3})} \right]$$

$$\lim_{x \rightarrow -\infty} \frac{x^3 \ominus}{x^2 \oplus} = -\infty$$

No H.A. !!!

$$f(x) = \frac{x^3}{x^2 - 1} \quad \begin{matrix} x=0 \\ y=0 \end{matrix}$$



Homework:

Vind the horizontal and vertical asymptotes of $y = \frac{x}{x-4}$.

V.A. $\lim_{x \rightarrow 4^+} \frac{x}{x-4} = \infty$

$\lim_{x \rightarrow 4^-} \frac{x}{x-4} = -\infty$

V.A. $x = 4$

H.A. $\lim_{x \rightarrow \infty} \frac{x}{x-4} = \lim_{x \rightarrow \infty} \frac{x}{x(1 - \frac{4}{x})} = 1$

$\lim_{x \rightarrow -\infty} \frac{x}{x-4} = 1$

H.A. $y = 1$

